

Improving tree selection for partial cutting through joint probability modelling of tree vigor and quality

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Abstract: Tree classification systems are generally designed to predict the supply of high-quality logs for the wood processing industries and to mark defective trees for removal with the aim of improving both the vigor and quality of future stands. Although these two objectives are generally inversely related, their joint consideration could enable the development of robust criteria for improved tree selection in partial cuttings of northern hardwood stands by targeting low-vigor (LV) trees of high quality (HQ). In this study, we used a copula approach to model the joint probability distribution of trees characterized by both vigor and quality, which accounts for the statistical dependence between the different classes of both classification systems. The relationships between the probability of occurrence of a tree in each joint category and tree diameter at breast height (DBH) were quite similar among the three studied species: yellow birch (*Betula alleghaniensis* Britt.), sugar maple (*Acer saccharum* Marsh.), and American beech (*Fagus grandifolia* Ehrh.). In particular, the probability of occurrence of LV–HQ trees was characterized by a parabolic curve with a maximum value attained at the mid-DBH range, whereas that of LV and low-quality (LQ) trees increased with increasing DBH. This suggests that instead of harvesting large and mostly LV–LQ trees, partial cutting should target smaller LV–HQ trees, so that the volume of harvested HQ trees would increase without compromising the silvicultural objective of improving the vigor of future stands.

Résumé : Les systèmes de classification des arbres sont généralement conçus pour prévoir l'approvisionnement en bois de haute qualité des industries de transformation et pour marquer les arbres défectueux à prélever dans le but d'augmenter la vigueur et la qualité des futurs peuplements. Bien que ces deux objectifs soient généralement contradictoires, leur prise en compte simultanée pourrait permettre de définir des critères robustes visant à améliorer la sélection des arbres à prélever lors de coupes partielles dans des peuplements de feuillus nordiques en ciblant les arbres de faible vigueur (FV) et de haute qualité (HQ). Dans cette étude, nous avons utilisé une approche par copule pour modéliser la distribution de la probabilité conjointe des arbres caractérisés par leur vigueur et leur qualité qui tient compte de la dépendance statistique entre les classes des deux systèmes de classification. La relation entre la probabilité qu'un arbre se situe dans chaque catégorie conjointe et son diamètre à hauteur de poitrine (DHP) était assez similaire pour les trois espèces étudiées : le bouleau jaune (*Betula alleghaniensis* Britt.), l'érable à sucre (*Acer saccharum* Marsh.) et le hêtre à grandes feuilles (*Fagus grandifolia* Ehrh.). De façon particulière, la probabilité d'occurrence des arbres FV–HQ était caractérisée par une courbe parabolique dont la valeur maximale était atteinte près du milieu de la gamme des DHP, alors que celle des arbres FV et de faible qualité augmentait avec le DHP. Ces résultats indiquent qu'au lieu de récolter les gros arbres qui sont souvent de faible vigueur et de faible qualité, les coupes partielles devraient cibler des arbres plus petits caractérisés par une faible vigueur et une haute qualité. Ainsi, on augmenterait la proportion du volume récolté constituée d'arbres de haute qualité sans compromettre l'objectif sylvicole d'améliorer la vigueur des futurs peuplements.

Introduction

Valuable tree species from the northern hardwood forests of North America have traditionally been harvested to supply the “appearance” wood products industries. The demand and market prices are particularly high for uniform pale wood from yellow birch (*Betula alleghaniensis* Britt.), sugar maple (*Acer saccharum* Marsh.), and to a lesser extent, American beech (*Fagus grandifolia* Ehrh.) (HMR 2011). However, the presence of discoloration or decay substantially reduces wood quality and value (Shigo 1967; Shortle and Cowling 1978; Sorz and Hietz 2008; Drouin et al. 2010) and has been linked to the presence of injuries on the tree stem, with dead branches being the most common example (Knoke 2003; Wernsdorfer et al. 2005; Belleville et al. 2011). Accordingly, stem injuries and other defects are the basic variables considered in stem quality and tree vigor classification systems.

Tree classification systems are generally designed to achieve two main objectives: (i) to predict the supply of high-quality logs

for the wood processing industries (financial objective) and (ii) to mark defective trees for removal so that the vigor and quality of future stands will be improved (silvicultural objective). In the majority of cases, tree quality classification systems evaluate the number and size of defects along the stem and assign a quality class based on criteria such as stem diameter, stem curvature, and clear bole length (Hanks 1976; Monger 1991). Although some of these criteria are visually assessed and therefore have a degree of subjectivity, such tree quality classifications have been confirmed as a useful indicator of log grade outturn (Fortin et al. 2009). In the northeastern hardwood forests of North America, several decades of management with the financial objective of supplying veneer mills and sawmills have resulted in depleted, low-vigor stands (Robitaille and Roberge 1981; Nyland 1992). Hence, the overriding emphasis on stem quality criteria (often referred to as “high-grading”) was deemed unsuitable for achieving the silvicultural objective of improving tree vigor and quality after partial cutting (Nyland 1998; Raymond et al. 2009). In other words, a stem classi-

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fication that focuses on quality only does not ensure the sustainable management of these hardwood stands.

The silvicultural objective could be achieved through the development of a classification system solely based on tree vigor (e.g., Boulet 2007). In 2005, as a response to public concerns about high-grading, forest managers of the province of Quebec, Canada, changed the standards in the application of selection cutting in hardwood stands on public lands with the clear objective of restoring the overall vigor of these stands by prioritizing the harvesting of low-vigor trees. These new standards were based on a tree vigor classification system. Vigor classes rely on the presence or absence of precursors of wood decay and other tree defects that are thought to affect survival probabilities in the upcoming 20 years (Boulet 2007). Vigor-based tree selection had an important negative impact on the appearance products value chain because low-vigor trees are often of lesser quality. Considering the low demand for pulpwood from hardwood species, the strict application of the tree vigor classification system markedly decreased the harvesting of northern hardwood forests. It is thus perceived as a major obstacle not only to the economic development of regions that depend on high quality wood industries, but also to the implementation of good forestry practices, because it contributes to the decline in forest harvesting and thereby prevents the application of sanitation cuts.

We propose that a combination of both quality- and vigor-based classification systems could help meet both silvicultural and financial objectives. A consistent harvesting strategy would then consist of marking and harvesting those low-vigor trees with the potential to produce high-quality wood before the point at which their quality starts to decline. Here, the major issue that remains is to determine when and under what conditions this high-quality to low-quality shift occurs. By analyzing the tree distribution among these combined vigor-quality classes, we expect to identify explanatory variables that can be related to growth dynamics. More specifically, the objectives of this study are thus (i) to model the joint probability distribution of trees characterized by their vigor and their quality, (ii) to determine tree, stand, and (or) regional characteristics that would maximize the occurrence of low-vigor trees that retained a high stem quality, and (iii) to propose practical criteria that could help select trees in partial cuttings to achieve both silvicultural and financial objectives. To do so, we used data from the provincial forest inventory in Quebec, Canada, for which quality and vigor classification systems were applied on the same trees. For the data analysis, we used a copula approach, which makes it possible to account for the statistical dependence between the different classes of the tree vigor classification system and those of the stem quality classification system.

Materials and methods

Data source

The information on the vigor and quality of the three species in this study was extracted from the temporary sample plots established by the Ministère des Ressources naturelles et de la Faune du Québec from 2003 to 2011. We selected only plots containing at least one tree of sugar maple, yellow birch, or American beech that was larger than 23.0 cm in diameter at breast height (DBH; 1.3 m above ground level), which is considered to be the minimum merchantable limit within the stem quality classification. These plots were well distributed across the province of Quebec south of 47°N, and some of them contained more than one considered species. Thus, the numbers of plots used for sugar maple, yellow birch, and American beech were 9700, 9103, and 2600, respectively, for a total of 16 113 plots. Within each of the circular 400 m² plots, all trees larger than 9.0 cm in diameter were measured and stratified into 2 cm diameter classes. In addition, the vigor and quality of all hardwood trees larger than 23.0 cm

were evaluated by trained observers according to distinct sets of criteria.

In addition to the usual diameter tally, the survey protocol also requires that a forest type be assigned to each plot according to a provincial ecological classification (Saucier et al. 1998). The assessment of the forest type is based on soil texture and deposition, as well as the presence or absence of some plant species groups. Two plots with the same forest type are thought to evolve towards the same late-successional vegetation, although their current development stages may be different. The data set encompassed seven forest types with their associated codes:

- the sugar maple – basswood forest type (FE2),
- the sugar maple – yellow birch forest type (FE3),
- the sugar maple – yellow birch – American beech forest type (FE4),
- the sugar maple – American hophornbeam forest type (FE5),
- the sugar maple – red oak forest type (FE6),
- the yellow birch – balsam fir – sugar maple forest type (MJ1),
- the eastern hemlock forest type (RT1).

Henceforth we will use the above three-character codes for the sake of clarity. A summary of the data set is provided in Table 1.

Vigor and quality classification systems

The vigor classification system separates trees into four categories that are assumed to be related to their probability of mortality during the next cutting cycle, i.e., a 20-year period (Boulet 2007). The distinction among classes is based on a visual examination of the presence and eventual severity of certain tree defects. These defects are grouped into eight categories: (1) conks and stromata, (2) cambium necrosis, (3) stem deformations and injuries, (4) stem base and root defects, (5) stem and bark cracks, (6) woodworms and sap wells, (7) crown decline, and (8) forks and pruning defects. An evaluation grid based on these various defects makes it possible to assign one of the four following classes to any tree: class M, trees that would likely die during the next cutting cycle; class S, trees with significant defects that do not compromise their survival in the medium term; class C, trees with some minor defects that are actively growing; and class R, trees without defects that should form the basis of the future stand. This classification system is currently used in Quebec to prioritize the trees to remove during partial cutting operations. In the absence of long-term monitoring results, a complete validation of the classification has not yet been carried out. However, growth-driven survival probabilities tended to confirm that trees from the lowest and highest vigor classes had different survival probabilities (Hartmann et al. 2008). In practice, classes M and S are often grouped together, as they represent trees with the lowest vigor and, consequently, those with the highest harvest priority. Similarly, trees that belong to classes C and R are often considered as high-vigor trees that should be left standing after partial cutting.

The quality classification system also differentiates trees into four categories with the aim of predicting both the lumber grade and volume recovery (Monger 1991). To this end, the best 3.7 m stem section located in the bottom 5 m of each tree is separated into four faces that are examined to detect the presence and the size of defects. Then, a class is attributed to each tree as a function of its DBH, the length of the clear (defect-free) bole on its third best face, an estimated percentage of wood rot, and the presence of stem sinuosity and curvature. The four classes, A to D in decreasing order of stem quality, are used to estimate the available wood volume for different types of wood processing industries (i.e., veneer, sawmill, or pulp and paper). Because each class was associated with a specific minimal diameter (>39, >33, >23, and >23 cm for classes A, B, C, and D, respectively), we reduced this classification to a two-class system by grouping classes A, B, and C together, while considering class D as the

Table 1. Summary of the data set used in this study (the percentages with respect to the total number of trees or the total number of plots are in parentheses).

Attributes	Yellow birch	Sugar maple	American beech
Tree-level information			
Number of trees	23 656	41 395	6865
Number of low vigor trees	8007 (33.8%)	15 964 (38.6%)	4334 (63.1%)
with high quality	5548 (23.5%)	11 409 (27.6%)	2251 (32.8%)
with low quality	2459 (10.4%)	4555 (11.0%)	2083 (30.3%)
Number of high vigor trees	15 649 (66.2%)	25 431 (61.4%)	2531 (36.9%)
with high quality	14 776 (62.5%)	24 225 (58.5%)	1825 (26.6%)
with low quality	873 (3.7%)	1206 (2.9%)	706 (10.3%)
DBH class median, minimum–maximum (cm)	24–118	24–96	24–68
Plot-level information			
Number of plots	9103	9700	2600
Basal area, minimum–maximum (m ² ·ha ⁻¹)	1.1–69.8	2.3–57.5	3.1–56.7
Stem density, minimum–maximum (stems·ha ⁻¹)	25–2975	41–2225	91–1823
Number of plots in each forest type			
FE2	245 (2.7%)	884 (9.1%)	211 (8.1%)
FE3	3309 (36.4%)	5656 (58.3%)	1543 (59.3%)
FE4	207 (2.3%)	330 (3.4%)	240 (9.2%)
FE5	54 (0.6%)	475 (4.9%)	125 (4.8%)
FE6	102 (1.1%)	532 (5.5%)	227 (8.7%)
MJ1	4877 (53.6%)	1652 (17.0%)	149 (5.7%)
RT1	309 (3.4%)	171 (1.8%)	105 (4.0%)

Note: Yellow birch, *Betula alleghaniensis*; sugar maple, *Acer saccharum*; American beech, *Fagus grandifolia*; FE2, sugar maple – basswood forest type; FE3, sugar maple – yellow birch forest type; FE4, sugar maple – yellow birch – American beech forest type; FE5, sugar maple – American hophornbeam forest type; FE6, sugar maple – red oak forest type; MJ1, yellow birch – balsam fir – sugar maple forest type; RT1, eastern hemlock forest type.

second group. This procedure facilitated the statistical analysis in which the group formed of classes A, B, and C was assumed to represent high-quality stems that included at least one sawlog, whereas class D represented low-quality stems, i.e., those with no sawlog potential.

Statistical analyses

Let us consider tree vigor ($y_{ij,1}$) and quality ($y_{ij,2}$) as two binary variables, so that $y_{ij,1} \sim \text{Bernoulli}(p_{ij,1})$ and $y_{ij,2} \sim \text{Bernoulli}(p_{ij,2})$, with $p_{ij,1}$ and $p_{ij,2}$ being the marginal probabilities for tree j in plot i to be of low vigor and of high quality, respectively. The term marginal probability refers to the probability of observing one response variable without knowledge of the other response variable. In other words, $p_{ij,1}$ is the probability of observing a low-vigor tree regardless of its quality class.

Under the assumption of independence, the joint probabilities of observing any of the four possible outcomes are as follows:

$$[1a] \quad \Pr(y_{ij,1} = 1, y_{ij,2} = 1) = p_{ij,1} p_{ij,2}$$

for low-vigor and high-quality trees,

$$[1b] \quad \Pr(y_{ij,1} = 0, y_{ij,2} = 1) = (1 - p_{ij,1}) p_{ij,2}$$

for high-vigor and high-quality trees,

$$[1c] \quad \Pr(y_{ij,1} = 1, y_{ij,2} = 0) = p_{ij,1} (1 - p_{ij,2})$$

for low-vigor and low-quality trees, and

$$[1d] \quad \Pr(y_{ij,1} = 0, y_{ij,2} = 0) = (1 - p_{ij,1})(1 - p_{ij,2})$$

for high-vigor and low-quality trees.

In the case of positive dependence, however, similar outcomes, i.e., $\mathbf{y}_{ij} = (1,1)^T$ and $\mathbf{y}_{ij} = (0,0)^T$, where T indicates a vector transposition, have higher probabilities of occurrence than $p_{ij,1}p_{ij,2}$, and

$(1 - p_{ij,1})(1 - p_{ij,2})$, whereas different outcomes, i.e., $\mathbf{y}_{ij} = (1,0)^T$ and $\mathbf{y}_{ij} = (0,1)^T$, have lower probabilities of occurrence. Conversely, in the case of negative dependence, different binary outcomes are more frequent and similar binary outcomes are less frequent than the products of the marginal probabilities.

The copula approach is one method that makes it possible to account for dependence between random variables. Copula models have recently been used for modelling joint distributions of continuous random variables (Wang et al. 2008; Kershaw et al. 2010; Wang et al. 2010), as well as in a context of spatial dependence among neighbouring trees (Fortin et al. 2013). This approach has been widely used in statistics for the modelling of dependent binary responses. It uses the properties of the marginal cumulative density functions (cdf) to create a joint cdf (Trevedi and Zimmer 2005, p.11):

$$[2] \quad F(y_{ij,1}, y_{ij,2}) = C(F_1(y_{ij,1}), F_2(y_{ij,2}); \theta)$$

where $F(y_{ij,1}, y_{ij,2})$ is the joint cdf of the multivariate distribution and $C(F_1(y_{ij,1}), F_2(y_{ij,2}); \theta)$ is the copula function based on parameter θ and the cumulative marginal distribution of the response variables. The marginal cdfs that compose the copula, i.e., $F_1(y_{ij,1})$ and $F_2(y_{ij,2})$ in eq. 2, are commonly referred to as the margins (Trevedi and Zimmer 2005, p. 3).

Many different bivariate copulas exist: the Farlie–Gumbel–Morgenstern copula, the Gaussian copula, the Clayton copula, and the Frank copula are well-known examples (Trevedi and Zimmer 2005, p. 16). In this study, the Gaussian copula was selected for its simplicity and its ability to model negative dependence.

The Gaussian copula is based on the probit function, i.e., the inverse cdf of the standard normal distribution. Considering that $F_1(y_{ij,1}) = p_{ij,1}$ and $F_2(y_{ij,2}) = p_{ij,2}$, the copula is defined as

$$[3] \quad C(p_{ij,1}, p_{ij,2}; \theta) = \Phi(\Phi^{-1}(p_{ij,1}), \Phi^{-1}(p_{ij,2}); \theta)$$

where $\Phi(\cdot)$ is the cdf of the standard normal distribution and $\Phi^{-1}(\cdot)$ is the probit function. Note that parameter θ is actually the correlation between the probit transformations of the marginal probabilities $p_{ij,1}$ and $p_{ij,2}$.

If the marginal binary regressions use the probit link function such that $\Phi^{-1}(p_{ij,1}) = \eta_{ij,1}$ and $\Phi^{-1}(p_{ij,2}) = \eta_{ij,2}$, the above Gaussian copula reduces to

$$[4] \quad C(p_{ij,1}, p_{ij,2}; \theta) = \int_{-\infty}^{\eta_{ij,1}} \int_{-\infty}^{\eta_{ij,2}} \frac{1}{2\pi\sqrt{1-\theta^2}} e^{\frac{-(w^2-2\theta wz+z^2)}{2\sqrt{1-\theta^2}}} dw dz$$

which is the joint probability of observing a low-vigor and high-quality tree, i.e., the vector $\mathbf{y}_{ij} = (1, 1)^T$. The probabilities associated with the other three outcomes are obtained by integrating on the balance of the probabilities as follows:

$$[5a] \quad \Pr(y_{ij,1} = 0, y_{ij,2} = 1) = \int_{\eta_{ij,1}}^{+\infty} \int_{-\infty}^{\eta_{ij,2}} \frac{1}{2\pi\sqrt{1-\theta^2}} e^{\frac{-(w^2-2\theta wz+z^2)}{2\sqrt{1-\theta^2}}} dw dz$$

for high-vigor and high-quality trees,

$$[5b] \quad \Pr(y_{ij,1} = 1, y_{ij,2} = 0) = \int_{-\infty}^{\eta_{ij,1}} \int_{\eta_{ij,2}}^{+\infty} \frac{1}{2\pi\sqrt{1-\theta^2}} e^{\frac{-(w^2-2\theta wz+z^2)}{2\sqrt{1-\theta^2}}} dw dz$$

for low-vigor and low-quality trees, and

$$[5c] \quad \Pr(y_{ij,1} = 0, y_{ij,2} = 0) = \int_{\eta_{ij,1}}^{+\infty} \int_{\eta_{ij,2}}^{+\infty} \frac{1}{2\pi\sqrt{1-\theta^2}} e^{\frac{-(w^2-2\theta wz+z^2)}{2\sqrt{1-\theta^2}}} dw dz$$

for high-vigor and low-quality trees.

The probit-transformed probabilities $\eta_{ij,1}$ and $\eta_{ij,2}$ decompose into linear terms such that $\eta_{ij,1} = \mathbf{x}_{ij}\boldsymbol{\beta}_1$ and $\eta_{ij,2} = \mathbf{x}_{ij}\boldsymbol{\beta}_2$, where $\mathbf{x}_{ij}$ is a vector of covariates and $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$ are two vectors of unknown population parameters. Moreover, to constrain the estimates of θ to the range $[-1, 1]$, this parameter can be re-expressed as $\theta = (e^\gamma - 1)/(e^\gamma + 1)$, with parameter γ being estimated in place of the original parameter.

The copula actually provides the joint probability of observing a particular vector $\mathbf{y}_{ij}$. The product of these joint probabilities provides the likelihood function for a particular data set. Maximizing the log-likelihood with respect to the model parameters yields the maximum likelihood estimates of $\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, and γ .

Model specifications

In a preliminary step, tree vigor and quality were independently modelled for each species to identify the best-fitting covariates. According to the aforementioned method, these responses were fitted using generalized linear models based on a probit link function. Pearson residuals were plotted against potential covariates to identify their effect and to ensure that there were no trends unaccounted for in the model. The contribution of each covariate to the model likelihood was assessed using Akaike's information criterion (AIC).

Once these independent regressions gave satisfactory results, the Gaussian copula was fitted for the three species independently, yielding the following models:

$$[6a] \quad \boldsymbol{\eta}_{ij} = \begin{bmatrix} \mathbf{x}_{ij}\boldsymbol{\beta}_1 \\ \mathbf{x}_{ij}\boldsymbol{\beta}_2 \end{bmatrix} = \begin{bmatrix} \beta_{1,0} + \beta_{1,1,v} + \beta_{1,2}dbh_{ij} + \beta_{1,3}dbh_{ij}^2 + \beta_{1,4}ba_i + \beta_{1,5}sd_i \\ \beta_{2,0} + \beta_{2,1,v} + \beta_{2,2}dbh_{ij} + \beta_{2,3}dbh_{ij}^2 + \beta_{2,4}ba_i + \beta_{2,5}sd_i \end{bmatrix}$$

for yellow birch,

$$[6b] \quad \boldsymbol{\eta}_{ij} = \begin{bmatrix} \mathbf{x}_{ij}\boldsymbol{\beta}_1 \\ \mathbf{x}_{ij}\boldsymbol{\beta}_2 \end{bmatrix} = \begin{bmatrix} \beta_{1,0} + \beta_{1,1,v} + \beta_{1,2}dbh_{ij} + \beta_{1,3}dbh_{ij}^2 \\ \beta_{2,0} + \beta_{2,1,v} + \beta_{2,2}dbh_{ij} + \beta_{2,3}dbh_{ij}^2 \end{bmatrix}$$

for sugar maple, and

$$[6c] \quad \boldsymbol{\eta}_{ij} = \begin{bmatrix} \mathbf{x}_{ij}\boldsymbol{\beta}_1 \\ \mathbf{x}_{ij}\boldsymbol{\beta}_2 \end{bmatrix} = \begin{bmatrix} \beta_{1,0} + \beta_{1,1,v} + \beta_{1,2}dbh_{ij} + \beta_{1,3}dbh_{ij}^2 + \beta_{1,4}ba_i + \beta_{1,5}sd_i \\ \beta_{2,0} + \beta_{2,1,v} + \beta_{2,2}dbh_{ij} + \beta_{2,3}dbh_{ij}^2 + \beta_{2,4}ba_i + \beta_{2,5}sd_i \end{bmatrix}$$

for American beech, where dbh_{ij} is the diameter at breast height (cm) of tree j in plot i , ba_i is the plot basal area ($\text{m}^2 \cdot \text{ha}^{-1}$), sd_i is the plot stem density ($\text{stems} \cdot \text{ha}^{-1}$), and v is the forest type index such that $v = 1, 2, 3, 4, 5$, and 6 for forest types FE3, FE4, FE5, FE6, MJ1, and RT1, respectively. Note that forest type FE2 served as the reference level. Plot basal area and stem density were omitted from the sugar maple model because these covariates did not significantly improve the maximum likelihood. Although DBH measurements were actually made by 2 cm diameter classes, for clarity we assumed that DBH was continuous and error free.

Model evaluation

The accuracy of the margins of the copula model was assessed using the area under the receiver operating characteristic (ROC) curve (cf. Lasko et al. 2005). This curve is produced by plotting the sensitivity of the model against one minus its specificity for any cut point between 0 and 1. For a particular cut point, the model sensitivity and specificity are calculated using the number of true positives and true negatives. The area under the ROC curve (AUC) is a common indicator of the accuracy of the model. A perfect model would yield an AUC value of 1.0, whereas random chance gives an AUC of 0.5 (Lasko et al. 2005). Models with an AUC ≥ 0.8 are considered to be accurate.

To justify the copula approach, we first compared the maximum log-likelihood of the independent regressions with the copula likelihood. In a second step, we also ran a score test. For each observation of the data set, the model provides four predicted probabilities, one for each possible outcome. These predicted probabilities were averaged across the data set and compared with observed frequencies. These differences between observed and predicted frequencies reduce to a 2×2 contingency table. The sum of the squared differences divided by the predicted frequencies is assumed to follow a χ^2 distribution with one degree of freedom. A significant probability indicates a lack of fit. This test was carried out both for the copula model and also under the assumption of independence, i.e., with the joint probabilities being calculated from the independent regressions, as shown in eqs. 1a, 1b, 1c, and 1d. The analyses were performed in the R statistical programming environment using the bprobit model available in the Zelig package (Imai et al. 2007). Nonparametric confidence intervals around the predicted probabilities were generated through Monte Carlo simulation. The Monte Carlo simulated probabilities were also used to produce confidence intervals around the difference between the probability of observing low-vigor and high-quality (LV-HQ) trees and that of observing low-vigor and low-quality (LV-LQ) trees. Because the alternative hypothesis was that observing LV-HQ trees had higher probabilities than observing LV-LQ trees,

Table 2. Fit statistics of the three models (the probability of observing a larger statistic value, $Pr > \chi^2$, is shown in parentheses).

Fit statistics	Yellow birch	Sugar maple	American beech
AUC			
Vigor margin	0.684	0.608	0.648
Quality margin	0.601	0.592	0.573
Maximum log-likelihood			
Based on independent regressions	-23 219.5	-43 159.5	-8875.5
Gaussian copula	-22 054.5	-40 930.9	-8747.1
Score test			
Based on independent regressions	2045.438 (<0.0001)	4145.072 (<0.0001)	230.108 (<0.0001)
Gaussian copula	0.237 (0.6261)	0.050 (0.8227)	0.004 (0.9499)

Note: Yellow birch, *Betula alleghaniensis*; sugar maple, *Acer saccharum*; American beech, *Fagus grandifolia*; AUC, area under the receiver operating characteristic (ROC) curve.

the hypothesis test was unilateral. Accordingly, we selected the maximum diameter for which this difference was larger than 0 with a 95% confidence level.

Results

The fit statistics of the different models are shown in Table 2. The AUC values were around 0.6, indicating that the accuracy was generally low. In all three cases, the copula model exhibited a higher log-likelihood value than the model based on the independent regressions. The maximum log-likelihood values increased by 1165.0 (5.0%), 2228.6 (5.2%), and 128.4 (1.4%) for the yellow birch, sugar maple, and American beech models, respectively. The score tests also indicated that only the models based on the independent regressions had important lack of fit, whereas the copula models did not.

The parameter estimates are shown in Tables 3, 4, and 5 for yellow birch, sugar maple, and American beech, respectively. According to the estimates of γ , parameter θ had estimated values of -0.60, -0.61, and -0.32 for yellow birch, sugar maple, and American beech, respectively. These estimated values indicated that the different binary outcomes had higher probabilities of occurrence than was predicted by the independent regressions. In other words, the associations between low quality (LQ) and low vigor (LV) and between high vigor (HV) and high quality (HQ) were stronger than expected. The differences between predictions from the copula model and from the independent regressions are shown in Fig. 1 for yellow birch trees. The predictions based on the independent regressions tended to overestimate the probabilities of observing HV and LQ trees, as well as LV and HQ trees (Figs. 1a, 1d), while underestimating the probabilities of observing HV and HQ trees and LV and LQ trees (Figs. 1b, 1c).

The relationships between the probability of occurrence of a tree in each joint category and tree DBH were quite similar among the three studied species (Fig. 2). For a given DBH, note that the sum of the probabilities of the four vigor-quality classes always equals 1. As a general rule, the predicted probability of occurrence of HV and LQ trees was low for any DBH, whereas that of HV and HQ trees decreased with increasing DBH. In contrast, the probability of occurrence of LV and HQ trees was characterized by a parabolic curve with a maximum value attained at mid-DBH range. Interestingly, the probability of occurrence of LV and LQ trees increased with increasing DBH and exceeded that of LV and HQ trees at DBH values slightly larger than the maximum value reached by the latter (Fig. 2). The confidence intervals of each curve generally increase with increasing DBH, which can be explained by the decreasing number of sample trees in larger diameter classes.

Although the general form of each probability curve is similar among species, the rate of development of these curves with DBH was different. Indeed, the maximum value of the probability of occurrence of LV and HQ trees was reached at around 64, 58, and 41 cm for yellow birch, sugar maple, and American beech, respec-

tively (Fig. 2). The confidence intervals around the differences between the probability of LV and HQ trees and that of LV and LQ trees made it possible to identify the maximum DBH for which the probability of the former was larger than the latter with a 95% confidence level (Table 6). For yellow birch, most maximum DBHs were greater than 70 cm, except for the FE4 forest type. These maximum DBHs were slightly smaller for sugar maple, with values ranging from 57 cm on the RT1 forest type to 68 cm on the FE5 forest type. For American beech, the maximum DBHs were around 40 cm on both the FE2 and FE3 forest types. For all other forest types, it was not possible to identify maximum DBHs because either the probability of observing LV and HQ trees was smaller than that of observing LV and LQ trees, or the 95% confidence level could not be determined.

The effects of plot basal area and stem density are shown in Fig. 3 for yellow birch and American beech, but not for sugar maple, as these two covariates were not part of the model for this species (Table 4). The probability of observing HV and HQ trees increased with basal area but decreased with stem density for both species (Fig. 3). LV and LQ trees tended to be less frequent with increasing basal area (Figs. 3a, 3b). On the other hand, the probability of observing these LV and LQ trees increased with stem density (Figs. 3c, 3d). Finally, LV and HQ American beech trees tended to increase in frequency with basal area but decreased with increasing stem density (Figs. 3b, 3d). The probability of observing LV and HQ yellow birch trees remained stable across the range of basal area and stem density (Figs. 3a, 3c). Predictions of other forest types showed similar trends. In fact, the major difference among the predictions of the different forest types was the width of the confidence intervals around the predictions, which was mainly explained by the different numbers of observations within each forest type.

Discussion

There was a high occurrence of HV trees in the diameter classes below the midrange DBH point of our sample plots for the three studied species, with most of them being classified as HQ. These plots are located in forest stands where development is usually driven by gap dynamics, with rare occurrences of severe stand-initiating disturbances (Seymour et al. 2002). As a consequence, it can be assumed that the vast majority of the plots were located in uneven-aged stands (Tyrrell and Crow 1994; Leak and Smith 1996; Guillemette et al. 2008). Therefore, these smaller diameter trees correspond to younger individuals that had not reached a position of dominance in the canopy. As these trees grow larger, they tend to progressively accumulate stem and (or) crown defects that result in a loss of vigor and also, for American beech, in a loss of stem quality. In fact, for all three species, the decrease of the HV-HQ curves is compensated for by an increase of the LV-HQ curve in the lower half of their respective DBH classes (Fig. 2).

To achieve both silvicultural and financial objectives, partial cutting should target LV-HQ trees. This is because the removal of

Table 3. Maximum likelihood parameter estimates of the yellow birch model (the standard errors of the parameter estimates appear in parentheses).

Parameter	Covariate	Vigor margin ($k = 1$)	Quality margin ($k = 2$)
$\beta_{k,0}$	Intercept	-2.0078 (0.1267)	1.2806 (0.1338)
$\beta_{k,1,1}$	Forest type FE3	1.7584×10^{-2} (6.6814×10^{-2})	-1.7109×10^{-2} (8.0785×10^{-2})
$\beta_{k,1,2}$	Forest type FE4	-7.5160×10^{-2} (9.7668×10^{-2})	-2.5199×10^{-1} (1.1127×10^{-1})
$\beta_{k,1,3}$	Forest type FE5	2.9601×10^{-2} (16.8010×10^{-2})	3.6667×10^{-1} (2.3849×10^{-1})
$\beta_{k,1,4}$	Forest type FE6	3.1323×10^{-1} (1.2563×10^{-1})	1.0123×10^{-1} (1.6152×10^{-1})
$\beta_{k,1,5}$	Forest type MJ1	-7.9901×10^{-2} (6.6121×10^{-2})	-8.4821×10^{-2} (7.9842×10^{-2})
$\beta_{k,1,6}$	Forest type RT1	6.6804×10^{-2} (8.4796×10^{-2})	6.9519×10^{-2} (10.5550×10^{-2})
$\beta_{k,2}$	dbh _{ij}	4.2002×10^{-2} (0.5013×10^{-2})	8.2821×10^{-3} (4.6419×10^{-2})
$\beta_{k,3}$	dbh _{ij} ²	1.5634×10^{-5} (5.8347×10^{-5})	-2.9793×10^{-4} (0.5067×10^{-4})
$\beta_{k,0}$	ba _i	2.2923×10^{-4} (0.3791×10^{-4})	-3.7095×10^{-4} (0.4304×10^{-4})
$\beta_{k,0}$	sd _i	-3.0890×10^{-3} (1.5087×10^{-3})	9.1807×10^{-3} (1.7748×10^{-3})
γ	Dependence	-1.3921 (0.0309)	

Note: Yellow birch, *Betula alleghaniensis*; FE3, sugar maple – yellow birch forest type; FE4, sugar maple – yellow birch – American beech forest type; FE5, sugar maple – American hornbeam forest type; FE6, sugar maple – red oak forest type; MJ1, yellow birch – balsam fir – sugar maple forest type; RT1, eastern hemlock forest type; ba_i, basal area; sd_i, stem density; dbh_{ij}, diameter at breast height (cm) of tree *j* in plot *i*.

Table 4. Maximum likelihood parameter estimates of the sugar maple model (the standard errors of the parameter estimates appear in parentheses).

Parameter	Covariate	Vigor margin ($k = 1$)	Quality margin ($k = 2$)
$\beta_{k,0}$	Intercept	-8.4939×10^{-1} (0.8922×10^{-1})	4.7347×10^{-2} (9.7350×10^{-2})
$\beta_{k,1,1}$	Forest type FE3	-9.3625×10^{-2} (2.1383×10^{-2})	2.2676×10^{-2} (2.6038×10^{-2})
$\beta_{k,1,2}$	Forest type FE4	-4.9287×10^{-1} (0.4240×10^{-1})	2.3036×10^{-1} (0.5225×10^{-1})
$\beta_{k,1,3}$	Forest type FE5	-1.7633×10^{-2} (3.3030×10^{-2})	1.8885×10^{-1} (0.4252×10^{-1})
$\beta_{k,1,4}$	Forest type FE6	1.0631×10^{-1} (0.3995×10^{-1})	-1.2225×10^{-1} (0.4731×10^{-1})
$\beta_{k,1,5}$	Forest type MJ1	0.9838×10^{-2} (3.0326×10^{-2})	-3.1260×10^{-1} (0.3492×10^{-1})
$\beta_{k,1,6}$	Forest type RT1	-1.7854×10^{-2} (7.2181×10^{-2})	2.1093×10^{-1} (0.9581×10^{-1})
$\beta_{k,2}$	dbh _{ij}	1.0240×10^{-2} (0.4671×10^{-2})	6.4746×10^{-2} (0.4940×10^{-2})
$\beta_{k,3}$	dbh _{ij} ²	2.4930×10^{-4} (0.5984×10^{-4})	-9.4397×10^{-4} (0.6092×10^{-4})
γ	Dependence	-1.4249 (0.0232)	

Note: Sugar maple, *Acer saccharum*; FE3, sugar maple – yellow birch forest type; FE4, sugar maple – yellow birch – American beech forest type; FE5, sugar maple – American hornbeam forest type; FE6, sugar maple – red oak forest type; MJ1, yellow birch – balsam fir – sugar maple forest type; RT1, eastern hemlock forest type; dbh_{ij}, diameter at breast height (cm) of tree *j* in plot *i*.

Table 5. Maximum likelihood parameter estimates of the American beech model (the standard errors of the parameter estimates appear in parentheses).

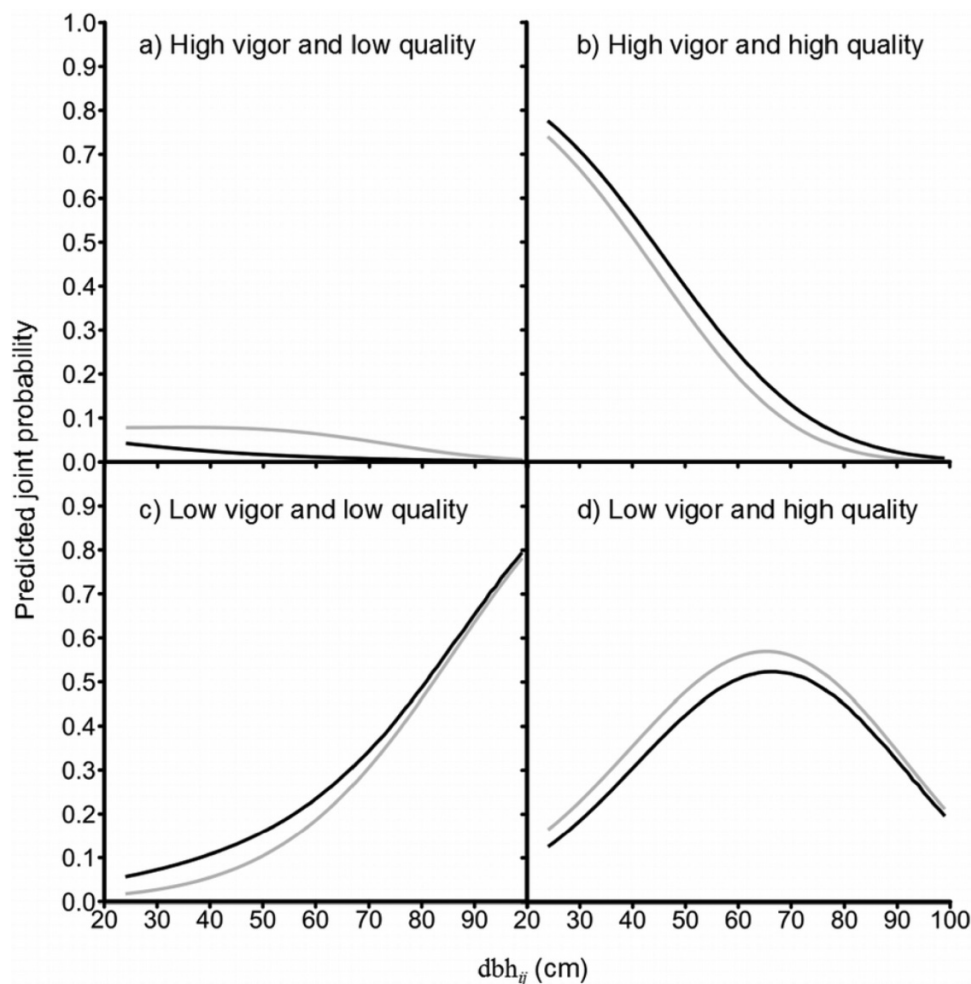
Parameter	Covariate	Vigor margin ($k = 1$)	Quality margin ($k = 2$)
$\beta_{k,0}$	Intercept	-2.2944 (0.3277)	0.4226×10^{-1} (3.0528×10^{-1})
$\beta_{k,1,1}$	Forest type FE3	8.8811×10^{-2} (6.2894×10^{-2})	-6.3346×10^{-2} (6.2111×10^{-2})
$\beta_{k,1,2}$	Forest type FE4	8.3077×10^{-2} (7.9595×10^{-2})	-4.0830×10^{-1} (0.7788×10^{-1})
$\beta_{k,1,3}$	Forest type FE5	-0.3507×10^{-1} (1.0498×10^{-1})	-0.9734×10^{-1} (1.0360×10^{-1})
$\beta_{k,1,4}$	Forest type FE6	7.8569×10^{-2} (8.3035×10^{-2})	-2.0113×10^{-1} (0.8146×10^{-1})
$\beta_{k,1,5}$	Forest type MJ1	-0.0436×10^{-1} (1.0366×10^{-1})	-1.3778×10^{-1} (1.0187×10^{-1})
$\beta_{k,1,6}$	Forest type RT1	-0.8433×10^{-1} (1.0894×10^{-1})	-0.4684×10^{-1} (1.0886×10^{-1})
$\beta_{k,2}$	dbh _{ij}	1.1239×10^{-1} (0.1813×10^{-1})	2.6606×10^{-2} (1.6493×10^{-2})
$\beta_{k,3}$	dbh _{ij} ²	-9.0285×10^{-4} (2.5710×10^{-4})	-5.9301×10^{-4} (2.2931×10^{-4})
$\beta_{k,0}$	ba _i	-5.6176×10^{-3} (2.6931×10^{-3})	1.2051×10^{-2} (0.2640×10^{-2})
$\beta_{k,0}$	sd _i	1.3065×10^{-4} (0.8848×10^{-4})	-3.6922×10^{-4} (0.8638×10^{-4})
γ	Dependence	-6.5771×10^{-1} (0.4173×10^{-1})	

Note: American beech, *Fagus grandifolia*; FE3, sugar maple – yellow birch forest type; FE4, sugar maple – yellow birch – American beech forest type; FE5, sugar maple – American hornbeam forest type; FE6, sugar maple – red oak forest type; MJ1, yellow birch – balsam fir – sugar maple forest type; RT1, eastern hemlock forest type; ba_i, basal area; sd_i, stem density; dbh_{ij}, diameter at breast height (cm) of tree *j* in plot *i*.

these trees will concentrate the development of the post-cut stand on high vigor trees, while the high stem quality of the harvested trees would allow the continued production of high-grade appearance products (Bédard and Majcen 2001). The parabolic shape of the LV-HQ probability curve for each species indicates that trees targeted for cutting are easier to find in a certain diameter range. The higher limit of this range can be set at the intersection of the LV-HQ and the LV-LQ curves (Table 6), because over this DBH value, the probability of obtaining low-quality trees is higher than that of obtaining high-quality trees.

When applying selection cuttings in Quebec, tree marking rules generally include a maximum diameter for the residual stand (Nyland 1998; Bohn et al. 2011), and all trees above this threshold are systematically marked. To harvest around one-third of the standing volume at each 15- to 25-year cutting cycle, additional trees are marked in all of the diameter class with a focus on LV trees. However, because of the historical legacy of high-grading in these northern hardwood forests, large LV-LQ trees are numerous (Fig. 2) and hence constitute a large proportion of the harvested volume. Based on the results of our model, we argue that it would

Fig. 1. Predicted joint probabilities as a function of diameter at breast height (cm) of tree j in plot i (dbh_{ij}) for yellow birch trees in the FE3 forest type (basal area (ba_i) = 24.5 $\text{m}^2\cdot\text{ha}^{-1}$; stand density (sd_i) = 607 stems $\cdot\text{ha}^{-1}$). Solid lines, predictions from the copula; shaded lines, predictions from the independent regressions.



be more profitable to replace the harvesting of a proportion of these large LV–LQ trees by an equivalent volume composed of smaller LV–HQ trees. By doing this, the volume of harvested HQ trees would increase without compromising the economic feasibility of the harvest operation, while also achieving the silvicultural objective of improving the vigor of future stands by applying selection cuttings.

These smaller LV–HQ trees seem to be easier to find within a diameter range that depends on species and forest types, as well as on stand basal area and density in the case of American beech. These diameter ranges seem to be particularly affected by the species, with a wider range and larger maximum diameter for yellow birch, followed by sugar maple and American beech. The vulnerability of American beech to bark injuries and to beech bark disease (Shigo 1967; Burns and Honkala 1990; Basham 1991; Houston 1994) is likely related to the narrower DBH range composed of trees around 40 cm that maximized the probability of occurrence of LV–HQ trees. This is consistent with the higher mortality rates of American beech observed beyond a DBH of 50 cm compared with both yellow birch and sugar maple (Fortin et al. 2008).

The present application of selection cutting, based on a maximum diameter for the residual stand, would imply that a high amount of smaller LV–LQ American beech trees will be perpetuated in future cutting cycles. This is an important issue for these forests as although there is still no consensus on the cause (Duchesne

and Ouimet 2009; Messier et al. 2011), the increasing presence of American beech at the sapling and pole stage has been well documented (Duchesne et al. 2005). Currently, American beech is considered as a low-quality species and as such is generally avoided by the appearance products industry. This has significantly reduced both the development of products derived from this species and the treatments applied to stands composed of a large proportion of American beech. Future silvicultural prescriptions and tree marking should take into account the different behaviour of American beech in terms of quality and vigor potential. By reducing the size of harvested trees, average stem quality would likely increase, with a consequent increase in demand. This could encourage forest managers to apply more silvicultural treatments in beech-dominated stands and thus help reduce, or at least control, the current upsurge in beech abundance (Duchesne et al. 2005). A strategy based on the production of small-diameter sawlogs appears to be the best potential solution.

Our suggestion to target smaller trees in partial cutting operations would have two important implications. First, many large trees that are currently harvested as part of normal forest operations would be left standing because their poor stem quality makes them unusable for the veneer and lumber industries. Although this is likely to be considered as a loss of wood volume by some forest managers, these large LV–LQ trees could, in turn, play important ecological roles in the residual stand such as maintaining a viable habitat for mosses, fungi, insects, and cavity-nesting

Fig. 2. Predicted joint probabilities and their 0.95 confidence intervals as a function of diameter at breast height (cm) of tree *j* in plot *i* (dbh_{ij}) for the three species in the FE3 forest type for given basal area (ba_i) and stand density (sd_i) conditions.

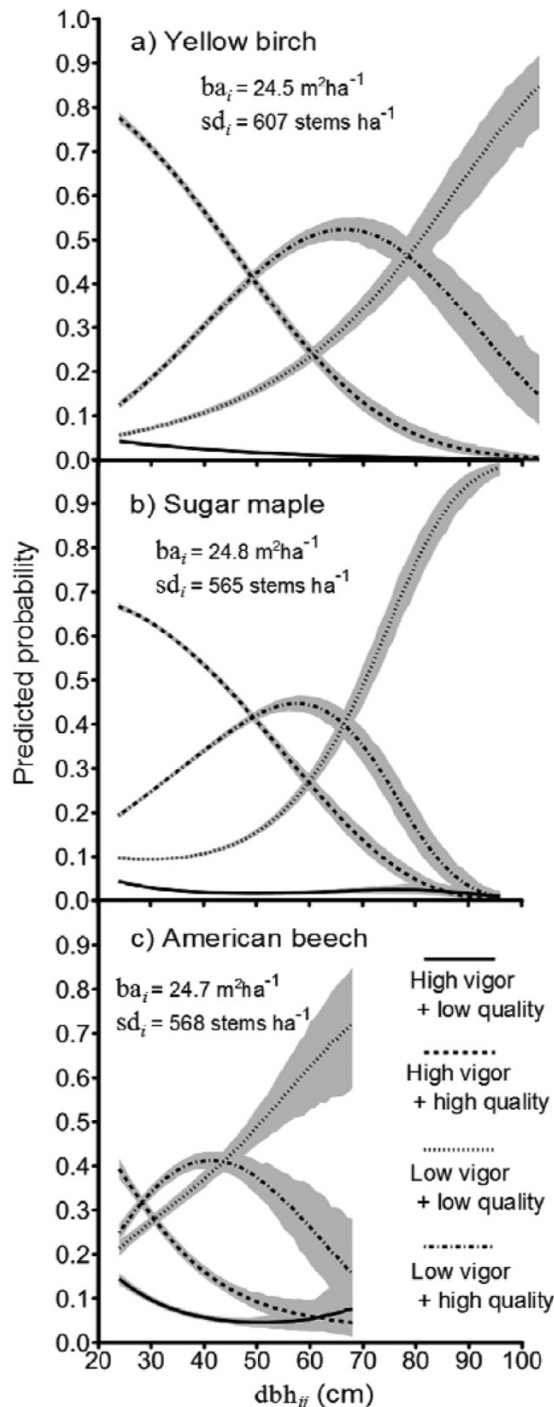


Table 6. Maximum diameter at breast height (DBH) for which the probability of observing a low-vigor and high-quality tree exceeds the probability of observing a low-vigor and low-quality tree (based on a unilateral 95% confidence interval with basal areas and stand densities set to the average of the data set).

Species	Forest type	Maximum DBH (cm)
Yellow birch	FE2	74
	FE3	75
	FE4	62
	FE5	78
	FE6	75
	MJ1	73
	RT1	76
Sugar maple	FE2	64
	FE3	65
	FE4	66
	FE5	68
	FE6	62
	MJ1	57
	RT1	67
American beech	FE2	40
	FE3	41
	FE4	None
	FE5	None
	FE6	None
	MJ1	None
	RT1	None

Note: Yellow birch, *Betula alleghaniensis*; sugar maple, *Acer saccharum*; American beech, *Fagus grandifolia*; FE2, sugar maple – basswood forest type; FE3, sugar maple – yellow birch forest type; FE4, sugar maple – yellow birch – American beech forest type; FE5, sugar maple – American hophornbeam forest type; FE6, sugar maple – red oak forest type; MJ1, yellow birch – balsam fir – sugar maple forest type; RT1, eastern hemlock forest type. “None” indicates there were no diameters for which the probability of low vigor and high quality exceeded that of low vigor and low quality with a confidence level of 95%.

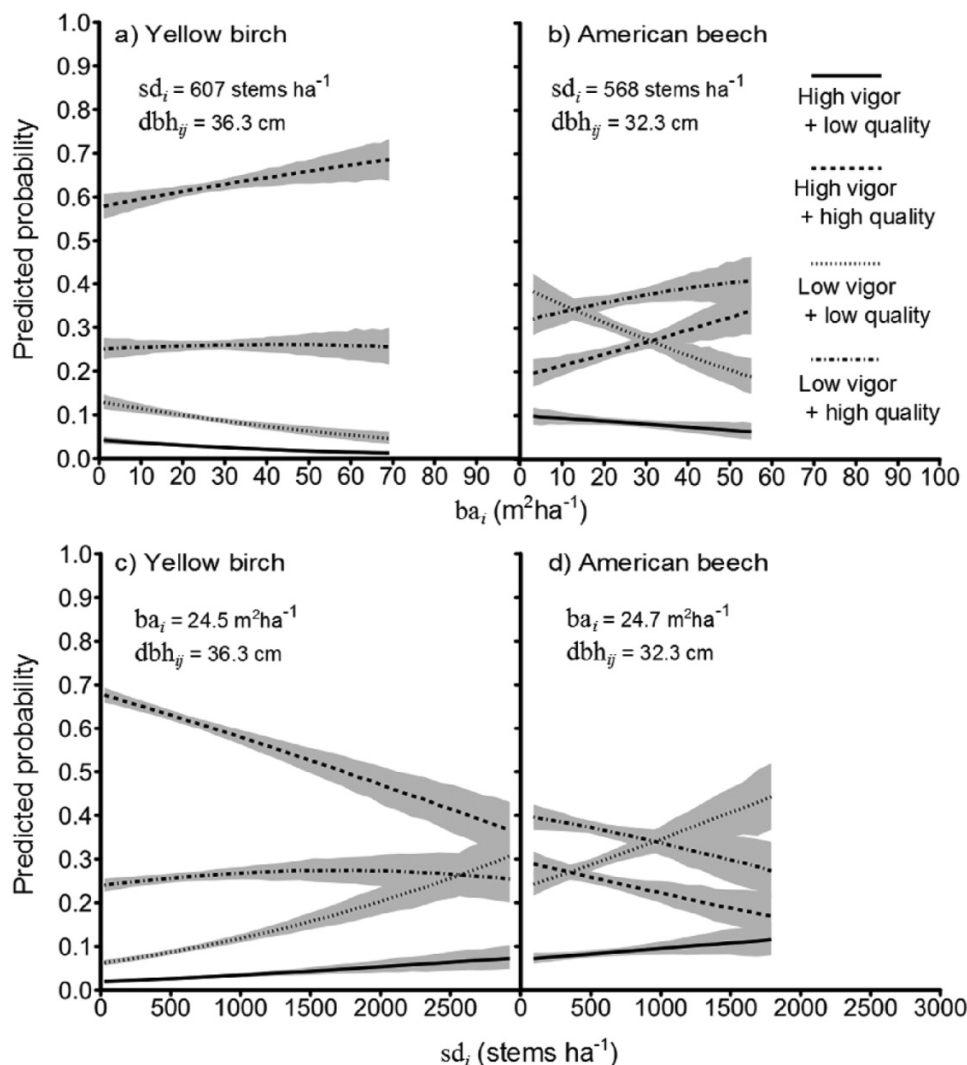
For all three species, the accuracy of the margins was rather low, indicating that there was still a large variability around the predictions that was not captured by the models. In fact, our margins rely on very few explanatory variables: tree DBH, plot stem density and basal area, as well as forest type. All these variables provide fairly crude indicators of the developmental history of the trees within a given stand, as they provide no information on previous management or disturbance patterns. Although we expected a higher accuracy, the AUC values that we observed are not surprising. Including additional indicators relating to stand disturbance and management could eventually help improve the accuracy of our model. This remains to be tested.

The Gaussian copula that we used made it possible to model the dependence between the quality and the vigor classes through a simple parameter. The copula has the advantage of providing both marginal predictions and joint predictions simultaneously. If the response of one of the margins is known, then this observation can be used to enhance the prediction of the other margin through the joint probability distribution (Fortin et al. 2013). In this sense, the copula approach offers more flexibility compared with a classic multinomial approach, which was an alternative option for analyzing our data.

Our model considers dependence at the tree level only. Actually, the copula takes into account the dependence between two tree characteristics: the vigor and the quality. It does not account for any dependence at the plot level. In other words, the responses of two trees located in the same plot are assumed to be independent. Taking this possible dependence into account is rather complex. An intuitive way would be to include plot random effects in the margins. However, coupling random effects and a copula in a

birds (Franklin et al. 1987; Busing 2005; Lowe et al. 2011; Larrieu and Cabanettes 2012). Second, the smaller diameter trees that would be harvested in place of the large LV–LQ trees, particularly beech, could require adjustments along the forest value chain (Wolfslehner et al. 2012). A key component of these adjustments may be the increased use of modern cant sawing techniques instead of traditional grade sawing, which is known to be more efficient for large, high-quality sawlogs (Steele 1984; Hamner et al. 2006).

Fig. 3. Predicted joint probabilities and their 0.95 confidence intervals as a function of basal area (ba_i) and stem density (sd_i) for yellow birch and American beech trees in the FE3 forest type for given mean size and stand conditions. dbh_{ij} , diameter at breast height (cm) of tree j in plot i .



single likelihood expression is likely to be complex, and to our knowledge, no study exists that has addressed this issue. Another option would be to consider both the plot- and tree-level dependences through hierarchical random effects. Nevertheless, such generalized linear mixed models are complex, and the computation of marginal predictions is not straightforward (Fortin 2013).

The previous comment on spatial dependence at the plot level is not limited to a single species. In fact, trees of all three studied species were found in some plots, and therefore it can reasonably be expected that the responses of one species is dependent on the responses of the other two. However, for the same aforementioned reasons, these possible dependences cannot be considered with the copula approach. Again, a complex generalized linear mixed model might be a way to consider this dependence, but this would have similar problems related to the computation of the marginal predictions. The comparison between the copula approach and a generalized linear mixed model approach remains to be investigated in the context of joint estimation of tree vigor and quality.

The sample sizes in our study increased the power of the statistical tests, especially the score tests. It could reasonably be argued that the differences between a standard approach with two independent regressions and the copula approach are small. In fact,

the largest differences were about 0.05 (Fig. 1). These differences are indeed small when dealing with a single stand. However, for forested areas such as those in Quebec, the rest of Canada, and the USA, these 0.05 differences in predicted probabilities are non-negligible. Although the total area treated through selection cutting has decreased from 31 000 to 9600 ha between 2004 and 2008 (Parent et al. 2012), it still represents a relatively large territory.

It is also important to highlight the fact that both stem quality and vigor are by no means binary variables. Here they were transformed into binary variables to help develop new approaches aimed at balancing the silvicultural and financial objectives underpinning the partial cutting strategy applied in the hardwood forests of Quebec. This simplification was helpful for statistical purposes, because it facilitated the modeling of the dependence between quality and vigor classes. In fact, the quality classification was not designed to be tractable from a statistical standpoint. The number of classes in this classification is dependent on tree DBH. For example, the best quality class is limited to trees with a minimum DBH of 39 cm, leaving two or three classes for smaller trees. Taking this fact into account in statistical models is complex, as different models would have to be fitted to trees in particular diameter class groups. We did not investigate this approach because of its complexity in relation to the potential

gain. Reducing both the quality and vigor to just two classes ensured statistical consistency and, moreover, represented a reasonable trade-off between practical applicability in the field and model complexity.

The simplicity of the proposed new system has an important practical application in tree marking, which itself is a binary process. Our main proposal is to add an estimate of quality to the vigor-based tree marking currently implemented for hardwood forests (Boulet 2007). Whereas trees containing no sawlogs (LQ) should be less valuable financially, there can still be large differences between trees containing only one small sawlog and those containing several large sawlogs (both HQ). In fact, even if only HQ trees are harvested, the general tree shape of hardwoods implies that there will always be a rather large proportion of lower quality “pulp” logs in the wood supply from northeastern hardwood forests (Fortin et al. 2009). In addition to the silvicultural solutions proposed here, it will be necessary to continue to pursue novel wood-processing solutions that could lead to higher added-value production from such logs (Lavoie and Stevanovic 2006; St-Pierre et al. 2013). The stated aim of restoring the hardwood forests will be best served by the application of integrated solutions along the entire spectrum of the value chain.

Our approach is based on the key assumption that current forest inventories reflect natural dynamics in the evolution of both vigor and quality. To fully test this, longitudinal measurements would be required. In other words, permanent-plot data in which vigor and quality classes were periodically measured on the same trees would make it possible to verify this assumption. However, these data are scarce, and in our context, no data set with longitudinal measurements of tree vigor was available because this tree classification system has only been recently implemented.

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